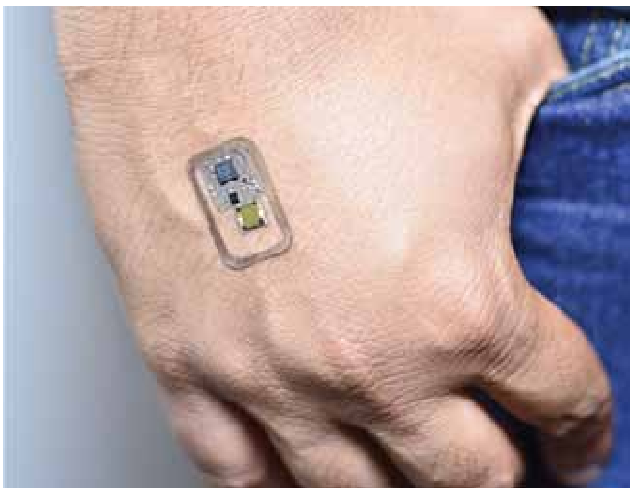
LAVO e-bike (Credit: https://www.electronicsforu.com)

for e-bikes. With the help of solar energy, LAVO's hybrid hydrogen battery operates to generate energy from the extraction of hydrogen from water. The hydrogen is then absorbed into a patented metal hydride, which converts it into battery power.

Battery-free wearable device measures nicotine exposure

Researchers from prominent Australian and the US research institutes have developed a battery-free, wearable device that helps e-cigarettes smokers to analyse in real time their



This flexible nicotine sensor attaches to skin, continuously measuring the wearer's exposure (Credit: www.acs.org)

nicotine consumption and measure the extent of acquiring cardiovascular and respiratory disorders. By choosing vanadium dioxide (VO_2) on a polyimide substrate as the basis for the sensor, it was shown that nicotine can bond covalently to a thin film of VO_2, thereby altering the film's conductivity depending on nicotine concentration, detecting the change in conductivity, and wirelessly transmitting it to a smartphone. The device can also help non-smokers to gauge the level of exposure to second-hand smoke.